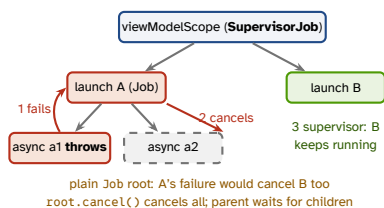
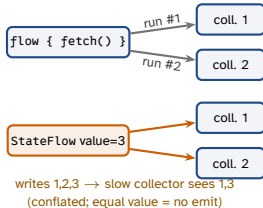


In one line: A coroutine is a **suspendable computation** launched in a **scope** whose Job tree gives **structured concurrency** (a parent waits for and cancels its children); **cancellation is cooperative**. Flow is a **cold** stream, **StateFlow/SharedFlow** are **hot** — the Kotlin twins of `async/await`, `Task`, `AsyncSequence` and `Combine` subjects.

Job tree — a child fails



Cold vs hot



Collecting on a lifecycle

	CREATED	STARTED	STOPPED (bg)	STARTED	DESTROY
repeatOnLifecycle		collect	cancelled	collect (new)	
launchWhenStarted		collect	paused*	collect	
launch { collect }	collects in background until DESTROYED				

Compose: `collectAsStateWithLifecycle()` = the green row
* paused, but the upstream stays alive — `launchWhenX` is deprecated

Coroutines — the mechanism

- `suspend fun` compiles to a **state machine** + a hidden Continuation param; suspending frees the thread (like `await`).
- `launch` → `Job` (fire-and-forget) · `async` → `Deferred<T>`, `await()`; `coroutineScope{}` waits for all children, rethrows the first failure; `supervisorScope{}` isolates them.
- **Scopes:** `viewModelScope` (`onCleared`), `lifecycleScope` (`onDestroy`), `rememberCoroutineScope` (leaves composition). `GlobalScope` = unstructured, leaks.
- **Dispatchers:** `Main` (UI; `Main.immediate` skips the re-post if already there) · `IO` (blocking, 64 threads or `#cores` if more) · `Default` (CPU, `#cores`). `withContext(IO) {}` hops and returns.
- **Cancellation is cooperative:** `CancellationException` at the next suspension point; CPU loops call `ensureActive()/yield()`. Blocking I/O is not interrupted.
- Errors: `launch` propagates up the Job tree to a `CoroutineExceptionHandler` on the **root** (ignored on children); `async` rethrows at `await()`.

Flow

- **Cold** `flow{}`: producer re-runs *per collector*; `Room/Retrofit` streams. `flowOn(IO)` moves *upstream* only; `.catch` sees *upstream* errors only.
- **StateFlow**: always a value, conflated, replays latest — screen state. **SharedFlow**: replay/buffer configurable — broadcasts.
- `stateIn(scope, WhileSubscribed(5_000), init)`: upstream survives the rotation gap, stops 5 s after the last collector. Eagerly/Lazily never stop.
- `flatMapLatest` cancels the previous inner flow (search); `combine` = latest of each (state), `zip` pairs by index.

Example — VM + safe collection

```
val ui: StateFlow<Ui> = repo.devices() // cold (Room)
    .map { Ui.Loaded(it) }
    .catch { emit(Ui.Error(it)) } // upstream only
    .stateIn(viewModelScope, WhileSubscribed(5_000), Ui.Loading)
private val _ev = Channel<Event>(Channel.BUFFERED)
val events = _ev.receiveAsFlow() // one-shot
fun refresh() = viewModelScope.launch {
    try { withContext(Dispatchers.IO) { repo.sync() } }
    catch (e: IOException) { _ev.send(Event.Offline) }
} // never catch(Exception)
// Activity / Fragment (viewLifecycleOwner):
lifecycleScope.launch { repeatOnLifecycle(STARTED) {
    vm.ui.collect { render(it) } } }
```

Coroutines ↔ Swift concurrency

Kotlin	Swift
<code>suspend fun</code>	<code>async func</code>
<code>launch {}</code> in a scope	<code>Task {}</code> (but <i>unstructured</i>)
<code>async/await</code> , <code>coroutineScope</code>	<code>async let</code> , <code>TaskGroup</code>
<code>Job.cancel()</code> · <code>ensureActive()</code>	<code>task.cancel()</code> · <code>Task.checkCancellation()</code>
<code>CancellationException</code>	<code>CancellationError</code>
<code>Dispatchers.Main</code> (runtime)	<code>@MainActor</code> (compile-time checked)
<code>withContext(IO)</code>	a nonisolated <code>async func</code> / <code>Task.detached actor</code>
<code>Mutex</code> · confined state	<code>AsyncSequence</code> · <code>AsyncStream(continuation)</code>
<code>Flow (cold)</code> · <code>callbackFlow</code>	<code>CurrentValueSubject</code> / <code>@Published</code>
<code>StateFlow</code> · <code>SharedFlow</code>	<code>PassthroughSubject</code>
<code>flatMapLatest</code> · <code>combine</code>	<code>switchToLatest</code> · <code>combineLatest</code>
<code>viewModelScope</code>	<code>Task cancelled in deinit</code>
<code>repeatOnLifecycle</code>	<code>.task { for await ... }</code>

Android vs iOS — platform

	Android	iOS
Entry	Activity, Service, Receiver, Provider; Intents route between apps	App + UIScenes; URL schemes
Links	App Links (assetLinks.json)	Universal Links (AASA)
Permiss.	runtime (API 23+); 2 denials = auto-deny; notifications from API 33	Info.plist string; one prompt, then Settings
Backgr.	WorkManager (persisted, constraints, periodic ≥ 15 min); fg service + notification; Doze	BGTaskScheduler (OS-timed), bg URLSession
Push	FCM : data vs notification; channels (26+)	APNs ; alert vs content-available
Devices	OEM skins, minSdk/targetSdk spread	few models, fast updates
Build	Gradle : build types × flavors, R8	Xcode configs, SPM
DI · DB	Hilt (Dagger, compile-time) · Room (checked SQL, Flow)	init injection · Core Data / SwiftData

Traps

- ! `catch (e: Exception)` swallows `CancellationException` — rethrow it or catch specific types.
- ! `StateFlow` for one-shot events re-fires after rotation; a `SharedFlow(replay=0)` drops events with no collector → `Channel`.
- ! Swift `Task{}` is *not* a child: nobody cancels it — store and cancel.

Remember

Scope owns Job owns children · cancel = cooperative · cold per collector, hot shared · collect from STARTED.